Aligning Metabolomic Networks via Graph Learning for Common Subnetwork Discovery

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Problem formulation

The mass spectrometry analysis includes two approach: **targeted** analysis and **untargeted** analysis.

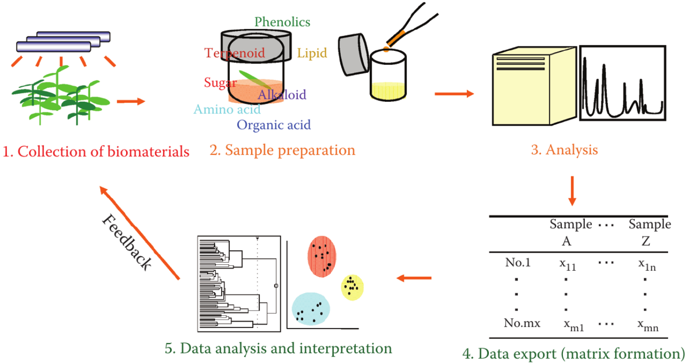

Figure 1: Metabolomics workflow.
Source: [?]

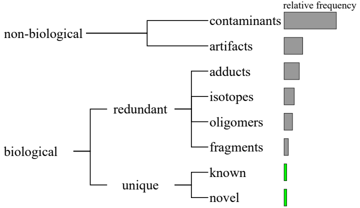

Figure 2: Untargeted composition.
Source: Adapted from [?]

Our objectives

Aligning metabolomic networks for filter Mass spectrometry data via Graph Neural Network models.

- Develop a pipeline for the analysis of metabolomic networks.
- Model metabolomic networks from mass spectrometry data.
- Implement GNN models to learn metabolite representations.
- Evaluate the effectiveness of the proposed GNN-based alignment approach.

Graph Neural Network (GNN)

It is a function $f : (A, X) \rightarrow Z$ that maps each node $v \in V$ of a graph $G = (V, E)$ to a low-dimensional embedding $z_v \in \mathbb{R}^d$ $(d \ll |V|)$.

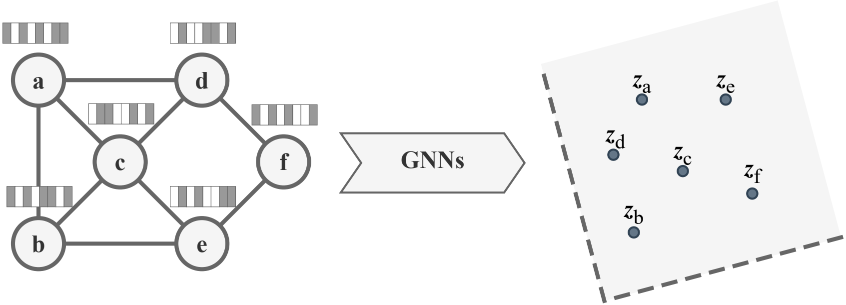

Figure 3: GNNs scheme.

Graph aligment problem

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs. The goal is to align their nodes through a mapping function $\pi : \mathbf{V_1} \rightarrow \mathbf{V_2}$, where each node $v \in V_1$ is mapped to its corresponding node $\pi(v) \in V_2$.

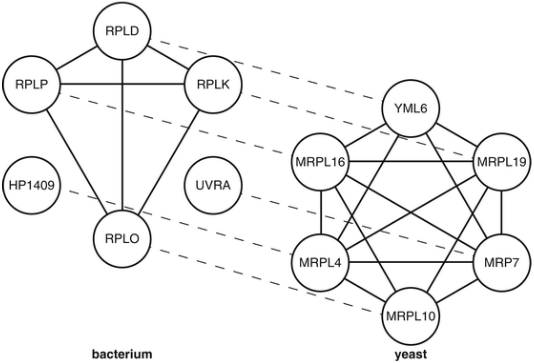

Figure 4: Node alignment.

Source: https://link.springer.com/rwe/10.1007/978-1-4419-9863-7_992

Base GNN model

T-GAE: Transferable Graph Autoencoder for Network Alignment.
(He, J., Kanatsoulis, C., & Ribeiro, A., 2024)

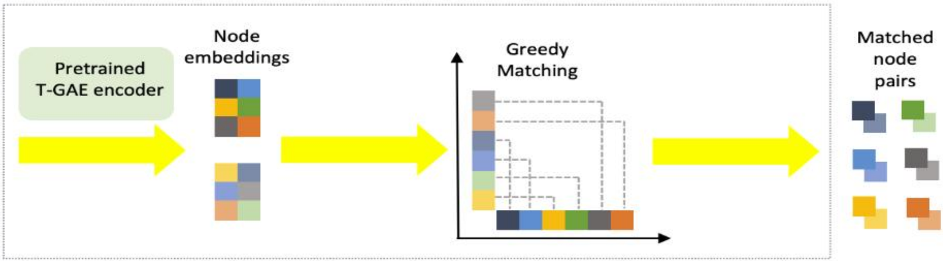

Figure 5: T-GAE encoder.

Method

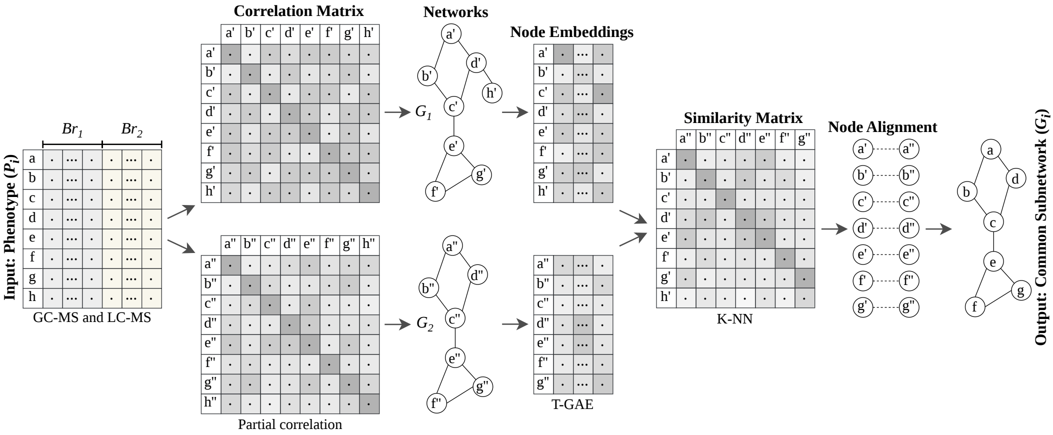

Figure 6: Our workflow.

Method overview

1. **Correlation estimation:** compute the partial correlation matrix $P = [\rho_{ij}]$ from mass espectrometry data.
2. **Network construction:** build a metabolic network using a correlation threshold ($|\rho_{ij}| \geq 0.5$), incorporate node features (like Rt, Mz, $Avg\ intensities$) and edge features (like $|\rho_{ij}|$, $\text{sign}(\rho_{ij})$).
3. **Graph representation learning:** learn node embeddings with the `T-GAE` architecture based on `GIN` and `GINE` layers.
4. **Network alignment:** identify metabolite correspondences using a `KNN`-based similarity matrix.

Hardware/software

The experiments were run on an AMD Ryzen Threadripper PRO 5955WX computer with 128 cores, 256 GB of RAM, 2x NVIDIA RTX A6000 with 48 GB of VRAM. The source code was implemented in Python with PyTorch Geometric.

Dataset

Experiments using the following datasets: Cacao, Leaf dataset (citar), Mentos datasets.

Experiments setup

Table 1: Experiments setup.

Features	Values
Node embedding	`T-GAE`
Architectures	`GIN`, `GINE`
Dimensions	16, 32, 64
Optimizer	Adam with $lr = 10^{-4}$, $weight\_decay = 5 \times 10^{-4}$
Early stopping	$patience = 10$
Epochs	100
Node features	Mz, Rt, $\log(Avg\ intensities)$, $Std\ intensities$, $Cv\ intensities$
Edge features	$\lvert\rho_{ij}\rvert$, $\mathrm{sign}(\rho_{ij})$, ρ_{ij}^2
Dataset	Cacao, Leaf, Mentos

Build metabolomic networks

Table 2: Cacao dataset.

Group	Subgroup	$\lvert V \rvert$	$\lvert E \rvert$	Density	Is connected?
SecoAmazonas	1	162	10268	0.787363	True
SecoAmazonas	2	162	8728	0.669274	True
SecoSanMartin	1	162	9771	0.749252	True
SecoSanMartin	2	162	9705	0.744191	True
SecoCusco	1	162	11297	0.866268	True
SecoCusco	2	162	8765	0.672111	True
FrescoAmazonas	1	162	9406	0.721264	True
FrescoAmazonas	2	162	9006	0.690591	True
FrescoSanMartin	1	162	9818	0.752856	True
FrescoSanMartin	2	162	9454	0.724944	True
FrescoCusco	1	162	10366	0.794878	True
FrescoCusco	2	162	9249	0.709225	True

Node embedding (T-GAE)

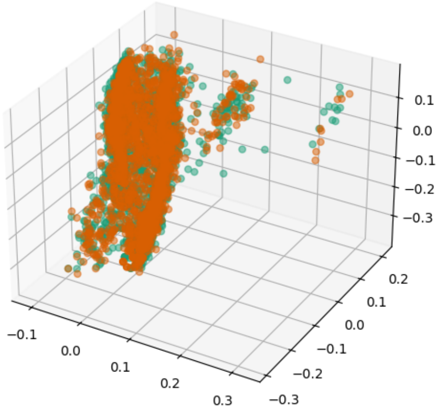

Figure 7: Node embedding (*secoSanMartin*).

Network alignment

Table 3: Node aligment (*secoSanMartin*).

ID 1	ID 2
13	13
19	252
27	378
31	1148
38	38
43	273
49	72
50	50
51	366
53	241
73	270
...	...
1825	1726

Table 4: Node filtered **133/2072** (*secoSanMartin*).

ID	Rt	Mz	Metabolite name
13	6.10	281.05	Unknown
38	6.16	182.10	Unknown
50	6.20	354.10	Unknown
128	6.47	119.20	Unknown
184	6.76	341.14	Unknown
261	7.23	371.06	Unknown
272	7.29	327.01	Unknown
314	7.61	57.02	DECANE; EI-B; MS
336	7.77	53.96	Unknown
339	7.81	328.01	Unknown
400	8.31	109.08	CITRONELLYL ACETATE...
402	8.31	109.13	BETA-HOMOCYCLOCITRAL...
407	8.37	401.11	Unknown

Comparison

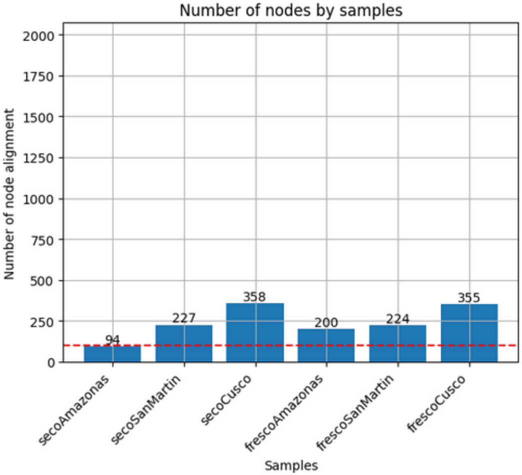

Figure 8: My image.

Conclusions

Our proposal, is a approach to the exploratory analysis of metabolic networks.

- Features shared by cacao beans within the same group are retained, reducing the number of features before identification is attempted.
- A core graph embedding is generated for each cacao bean, providing a compact representation that can be used as input to a classifier for subsequent cacao bean identification.
- The alignment workflow enables samples acquired on different days to be integrated into a single dataset. This approach mitigates known batch effects, such as retention-time shifts, while the graph embedding favors correlations among signals rather than depending on their absolute intensity values.

Thanks!

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`https://github.com/win7`

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